

UTS Pemrograman Dasar – Jawaban Lengkap

Tema: Data, Game, dan Analisis Otomatis

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Kelas: Teknologi Informasi 2

Dosen: Rintis Mardika Sunarto

Repository:

[https://github.com/Rasya2445/UTS_PemrogramanDasar_MuhammadFirasyaAlfathSuynto_2025806017.git]

1. Struktur Repository GitHub

UTS_PemrogramanDasar_MuhammadFirasyaAlfathSuyanto_2025806017/

```
|
|_ C/
|   |_ soal1_data_mahasiswa/
|       |_ linked_list.c
|       |_ linked_list.h
|       |_ main.c
|       |_ data_mahasiswa.csv
|       |_ README.md
|
|   |_ soal4_csv_json/
|       |_ data_mahasiswa.csv
|       |_ data_mahasiswa.json
|       |_ data_mahasiswa.py
|       |_ README.md
|
|_ Python/
    |_ soal2_game_guess/
        |_ game.py
        |_ main.py
        |_ scoreboard.py
        |_ scores.json
        |_ README.md
    |_ soal3_text_analyzer/
        |_ analyzer.py
        |_ main.py
        |_ utils.py
        |_ input.txt
        |_ report.txt
        |_ README.md
    |_ soal4_csv_json/
        |_ convert.py
```

```

|       |----- data_mahasiswa.csv
|       |----- data_mahasiswa.json
|       |----- README.md
|
|----- docs/
|       |----- output_soal1.png
|       |----- output_soal2.png
|       |----- output_soal3.png
|       |----- output_soal4.png
|       |----- output_soal4_c.png
|
|----- README.md

```

2. Soal 1 – Sistem Data Mahasiswa Terintegrasi (C)

File: C/mahasiswa.c

Linked_list.h:

```
#ifndef LINKED_LIST_H
```

```
#define LINKED_LIST_H
```

```
typedef struct Mahasiswa {
```

```
    char nama[100];
```

```
    char nim[20];
```

```
    float tugas;
```

```
    float uts;
```

```
    float uas;
```

```
    float akhir;
```

```
    char mutu;
```

```
    struct Mahasiswa *next;
```

```
} Mahasiswa;
```

```
// Fungsi utama
```

```
void addMahasiswa(Mahasiswa **head);
```

```

void tampilMahasiswa(Mahasiswa *head);

void cariMahasiswa(Mahasiswa *head);

void hapusMahasiswa(Mahasiswa **head);

void simpanCSV(Mahasiswa *head);

void freeList(Mahasiswa *head);


// Fungsi bantu

float hitungNilaiAkhir(float tugas, float uts, float uas);

char hitungMutu(float akhir);


#endif

```

Linked_list.c:

```

#include <stdio.h>

#include <stdlib.h>

#include <string.h>

#include "linked_list.h"


float hitungNilaiAkhir(float tugas, float uts, float uas) {

    return (0.3f * tugas) + (0.3f * uts) + (0.4f * uas);

}


char hitungMutu(float akhir) {

    if (akhir >= 85)

        return 'A';

    else if (akhir >= 70)

```

```
        return 'B';
    else if (akhir >= 60)
        return 'C';
    else if (akhir >= 50)
        return 'D';
    else
        return 'E';
}
```

```
void addMahasiswa(Mahasiswa **head) {
    Mahasiswa *baru = (Mahasiswa *)malloc(sizeof(Mahasiswa));

    if (baru == NULL) {
        printf("Gagal alokasi memori!\n");
        return;
    }
```

```
    printf("Nama      : ");
    getchar(); // membersihkan newline
    fgets(baru->nama, sizeof(baru->nama), stdin);
    baru->nama[strcspn(baru->nama, "\n")] = '\0';
```

```
    printf("NIM      : ");
    scanf("%s", baru->nim);
```

```
    printf("Nilai Tugas : ");
```

```
scanf("%f", &baru->tugas);
```

```
printf("Nilai UTS  : ");
```

```
scanf("%f", &baru->uts);
```

```
printf("Nilai UAS  : ");
```

```
scanf("%f", &baru->uas);
```

```
baru->akhir = hitungNilaiAkhir(
```

```
    baru->tugas,
```

```
    baru->uts,
```

```
    baru->uas
```

```
);
```

```
baru->mutu = hitungMutu(baru->akhir);
```

```
baru->next = *head;
```

```
*head = baru;
```

```
printf("\nData mahasiswa berhasil ditambahkan!\n");
```

```
}
```

```
void tampilMahasiswa(Mahasiswa *head) {
```

```
    if (head == NULL) {
```

```
        printf("\nData masih kosong.\n");
```

```
        return;
```

```
}
```

```
printf("\n=====
=====\\n");
```

```
printf("%-20s %-15s %-8s %-8s %-8s %-12s %-5s\\n",
      "Nama", "NIM", "Tugas", "UTS", "UAS", "Nilai Akhir", "Mutu");
```

```
printf("=====
=====\\n");
```

```
Mahasiswa *temp = head;
```

```
while (temp != NULL) {
```

```
printf("%-20s %-15s %-8.2f %-8.2f %-8.2f %-12.2f %-5c\\n",
```

```
temp->nama,
```

```
temp->nim,
```

```
temp->tugas,
```

```
temp->uts,
```

```
temp->uas,
```

```
temp->akhir,
```

```
temp->mutu);
```

```
temp = temp->next;
```

```
}
```

```
printf("=====
=====\\n");
```

```
}
```

```

void cariMahasiswa(Mahasiswa *head) {

    char nim[20];

    printf("Masukkan NIM yang dicari: ");

    scanf("%s", nim);

    Mahasiswa *temp = head;

    while (temp != NULL) {

        if (strcmp(temp->nim, nim) == 0) {

            printf("\nData ditemukan:\n");

            printf("Nama      : %s\n", temp->nama);

            printf("NIM       : %s\n", temp->nim);

            printf("Nilai Akhir : %.2f\n", temp->akhir);

            printf("Mutu       : %c\n", temp->mutu);

            return;

        }

        temp = temp->next;

    }

    printf("Data tidak ditemukan.\n");

}

```

```

void hapusMahasiswa(Mahasiswa **head) {

    char nim[20];

    printf("Masukkan NIM yang akan dihapus: ");

    scanf("%s", nim);

```

```
Mahasiswa *temp = *head;
```

```
Mahasiswa *prev = NULL;
```

```
while (temp != NULL && strcmp(temp->nim, nim) != 0) {
```

```
    prev = temp;
```

```
    temp = temp->next;
```

```
}
```

```
if (temp == NULL) {
```

```
    printf("Data tidak ditemukan.\n");
```

```
    return;
```

```
}
```

```
if (prev == NULL) {
```

```
    *head = temp->next;
```

```
} else {
```

```
    prev->next = temp->next;
```

```
}
```

```
free(temp);
```

```
printf("Data berhasil dihapus.\n");
```

```
}
```

```
void simpanCSV(Mahasiswa *head) {
```

```
    FILE *fp = fopen("data_mahasiswa.csv", "w");
```



```
if (fp == NULL) {  
    printf("Gagal membuat file CSV.\n");  
    return;  
}  
  
fprintf(fp, "Nama,NIM,Tugas,UTS,UAS,NilaiAkhir,Mutu\n");  
  
Mahasiswa *temp = head;  
while (temp != NULL) {  
    fprintf(fp,  
        "%s,%s,%.2f,%.2f,%.2f,%.2f,%c\n",  
        temp->nama,  
        temp->nim,  
        temp->tugas,  
        temp->uts,  
        temp->uas,  
        temp->akhir,  
        temp->mutu);  
    temp = temp->next;  
}  
  
fclose(fp);  
printf("Data berhasil disimpan ke data_mahasiswa.csv\n");  
}
```

```

void freeList(Mahasiswa *head) {
    Mahasiswa *temp;

    while (head != NULL) {
        temp = head;
        head = head->next;
        free(temp);
    }
}

```

Main.c:

```

#include <stdio.h>
#include "linked_list.h"

int main() {
    Mahasiswa *head = NULL;
    int pilihan;

    do {
        printf("\n===== MENU DATA MAHASISWA =====\n");
        printf("1. Tambah Mahasiswa\n");
        printf("2. Tampilkan Mahasiswa\n");
        printf("3. Cari Mahasiswa\n");
        printf("4. Hapus Mahasiswa\n");
        printf("5. Simpan ke CSV\n");
        printf("0. Keluar\n");
    } while (pilihan != 0);
}

```

```
printf("Pilih menu: ");  
scanf("%d", &pilihan);  
  
switch (pilihan) {  
    case 1:  
        addMahasiswa(&head);  
        break;  
    case 2:  
        tampilMahasiswa(head);  
        break;  
    case 3:  
        cariMahasiswa(head);  
        break;  
    case 4:  
        hapusMahasiswa(&head);  
        break;  
    case 5:  
        simpanCSV(head);  
        break;  
    case 0:  
        freeList(head);  
        printf("Program selesai.\n");  
        break;  
    default:  
        printf("Pilihan tidak valid!\n");  
}
```

```

    } while (pilihan != 0);

    return 0;
}

```

Instruksi run:

Cd C\soal1_data_mahasiswa

gcc main.c linked_list.c -o program

Hasil output:

```

C:\Users\Lenovo\Documents>
===== MENU DATA MAHASISWA =====
1. Tambah Mahasiswa
2. Tampilkan Mahasiswa
3. Cari Mahasiswa
4. Hapus Mahasiswa
5. Simpan ke CSV
0. Keluar
Pilih menu: 1
Nama      : Andi
NIM       : 2024806033
Nilai Tugas : 80
Nilai UTS  : 85
Nilai UAS  : 90
Data mahasiswa berhasil ditambahkan!
===== MENU DATA MAHASISWA =====
1. Tambah Mahasiswa
2. Tampilkan Mahasiswa
3. Cari Mahasiswa
4. Hapus Mahasiswa
5. Simpan ke CSV
0. Keluar
Pilih menu: 5
Data berhasil disimpan ke data_mahasiswa.csv
===== MENU DATA MAHASISWA =====
1. Tambah Mahasiswa
2. Tampilkan Mahasiswa
3. Cari Mahasiswa
4. Hapus Mahasiswa
5. Simpan ke CSV
0. Keluar
Pilih menu: |

```

3. Soal 2 – Guess Battle (Python)

File: Python/game.py

game.py:

```

import random

def play_game(level):
    if level == 1:
        max_number = 10
        attempts = 3
    elif level == 2:
        max_number = 50
        attempts = 5
    elif level == 3:
        max_number = 100
        attempts = 7
    else:
        print("Level tidak valid!")
        return 0

    secret_number = random.randint(1, max_number)

    print(f"\n=== LEVEL {level} ===")
    print(f"Tebak angka antara 1 sampai {max_number}")
    print(f>Kesempatan: {attempts} kali")

    for i in range(attempts):
        try:
            guess = int(input(f>Tebakan ke-{i+1}: "))
        except ValueError:
            print("Input harus berupa angka!")
            continue

        if guess == secret_number:
            score = (attempts - i) * 10
            print(f"Benar! Angka yang dicari adalah {secret_number}")
            print(f"Skor Anda: {score}")
            return score
        elif guess < secret_number:
            print("Terlalu kecil!")
        else:
            print("Terlalu besar!")

    print(f"Game over! Angka yang benar adalah {secret_number}")
    return 0

scoreboard.py:
import json
import os

FILE_SCORE = "scores.json"

def load_scores():
    if not os.path.exists(FILE_SCORE):

```

```

        return []

    try:
        with open(FILE_SCORE, "r") as file:
            return json.load(file)
    except:
        return []

def save_score(name, score):
    scores = load_scores()
    scores.append({
        "name": name,
        "score": score
    })

    scores.sort(key=lambda x: x["score"], reverse=True)

    with open(FILE_SCORE, "w") as file:
        json.dump(scores, file, indent=4)

def show_top5():
    scores = load_scores()

    if not scores:
        print("\nBelum ada skor.")
        return

    print("\n=== TOP 5 SCORE ===")
    for i, data in enumerate(scores[:5], start=1):
        print(f"{i}. {data['name']} - {data['score']} pts")

main.py:
from game import play_game
from scoreboard import save_score, show_top5

def main():
    print("==== GUESS BATTLE ====")

    name = input("Masukkan nama pemain: ")

    total_score = 0

    for level in range(1, 4):
        score = play_game(level)
        total_score += score

    if score == 0:
        print("Anda gagal pada level ini.")
        break

```

```
print(f"\nTotal skor {name}: {total_score}")
```

```
save_score(name, total_score)  
show_top5()
```

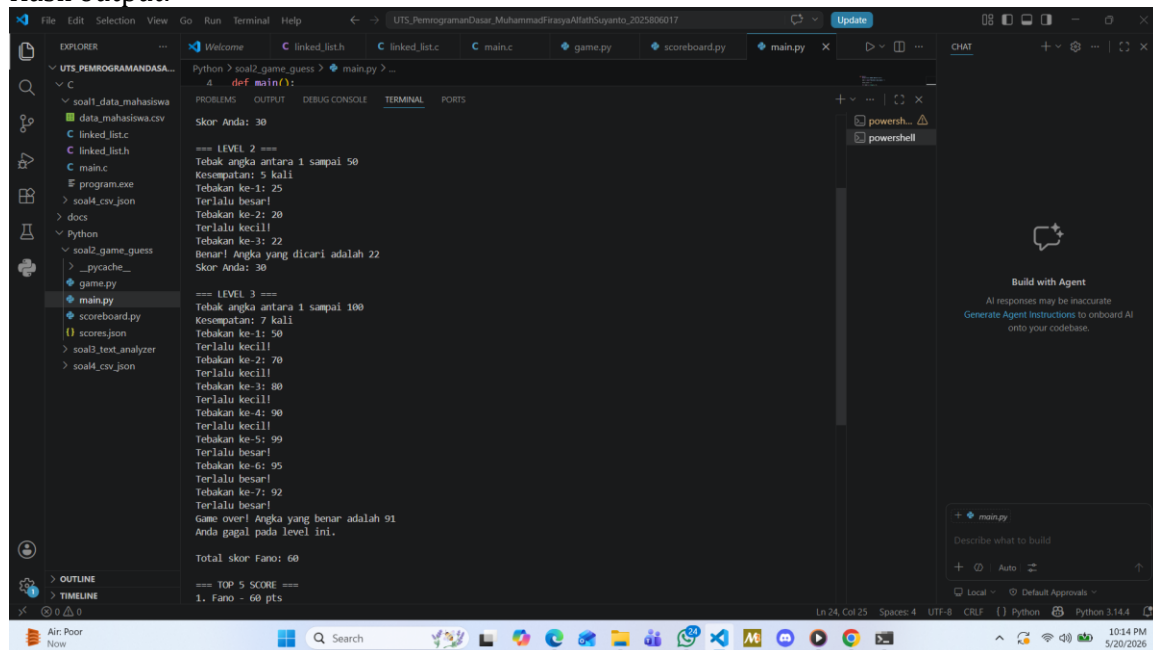
```
if __name__ == "__main__":  
    main()
```

Instruksi run:

Cd Python\soal2_game_guess

python main.py

Hasil output:



```
Python > soal2_game_guess > main.py > ...  
4 def main():  
    Skor Anda: 30  
  
    === LEVEL 2 ===  
    Tebak angka antara 1 sampai 50  
    Kesempatan: 5 kali  
    Tebakan ke-1: 25  
    Terlalu besar!  
    Tebakan ke-2: 20  
    Terlalu kecil!  
    Tebakan ke-3: 22  
    Benar! Angka yang dicari adalah 22  
    Skor Anda: 30  
  
    === LEVEL 3 ===  
    Tebak angka antara 1 sampai 100  
    Kesempatan: 7 kali  
    Tebakan ke-1: 50  
    Terlalu kecil!  
    Tebakan ke-2: 70  
    Terlalu kecil!  
    Tebakan ke-3: 80  
    Terlalu kecil!  
    Tebakan ke-4: 90  
    Terlalu kecil!  
    Tebakan ke-5: 99  
    Terlalu besar!  
    Tebakan ke-6: 95  
    Terlalu besar!  
    Tebakan ke-7: 92  
    Terlalu besar!  
    Game over! Angka yang benar adalah 91  
    Anda gagal pada level ini.  
  
    Total skor Fano: 60  
  
    === TOP 5 SCORE ===  
    1. Fano - 60 pts
```

4. README.md

UTS Pemrograman Dasar

Identitas Mahasiswa

- Nama : Muhammad Firasya Alfath Suyanto

- NIM : 2025806017

- Kelas: [Teknologi Informasi 2]

- Dosen: [Rintis Mardika Sunarto]

- Repository:

[https://github.com/Rasya2445/UTS_PemrogramanDasar_MuhammadFirasyaAlfathSuynto_2025806017.git]

Deskripsi Project

Repository ini berisi jawaban Ujian Tengah Semester (UTS) Pemrograman Dasar.

Project terdiri dari 4 soal utama:

1. Sistem Data Mahasiswa Terintegrasi (Bahasa C)
2. Guess Battle Game (Python)
3. Analisis Teks Otomatis (Python)
4. Konversi CSV ke JSON

Struktur Folder

```text

UTS\_PemrogramanDasar\_MuhammadFirasyaAlfathSuyanto\_2025806017/

```
|
|_ C/
| |_ soal1_data_mahasiswa/
| |_ linked_list.c
| |_ linked_list.h
| |_ main.c
| |_ data_mahasiswa.csv
| |_ README.md
| |_ soal4_csv_json/
| |_ data_mahasiswa.csv
| |_ data_mahasiswa.json
| |_ data_mahasiswa.py
| |_ README.md
|
|_ Python/
 |_ soal2_game_guess/
 |_ game.py
 |_ main.py
 |_ scoreboard.py
 |_ scores.json
 |_ README.md
 |_ soal3_text_analyzer/
 |_ analyzer.py
 |_ main.py
 |_ utils.py
 |_ input.txt
 |_ report.txt
 |_ README.md
 |_ soal4_csv_json/
 |_ convert.py
```



```
├── data_mahasiswa.csv
├── data_mahasiswa.json
├── README.md
├── docs/
│ ├── output_soal1.png
│ ├── output_soal2.png
│ ├── output_soal3.png
│ ├── output_soal4.png
│ └── output_soal4_c.png
└── README.md
```

## 5. Perintah GitHub

```
git init
git add .
git commit -m "feat: menyelesaikan seluruh soal UTS"
git branch -M main
git remote add origin
https://github.com/username/UTS_PemrogramanDasar_Nama_NIM.git
git push -u origin main
```

## 6. Soal 3 – Analisis Teks Otomatis (Python Advanced)

File: Python/analyzer.py

Utils.py:

```
import re
```

```
from collections import Counter
```

```
def read_file(filename):
```

```
 """Membaca isi file teks."""
```

```
 with open(filename, "r", encoding="utf-8") as file:
```

```
 return file.read()
```

```
def clean_words(text):
```

```
 """Membersihkan teks dan mengubah menjadi list kata."""
```

```
text = text.lower()

words = re.findall(r'\b[a-zA-Z]+\b', text)

return words
```

```
def count_lines(text):

 """Menghitung jumlah baris."""

 return len(text.splitlines())
```

```
def count_words(words):

 """Menghitung jumlah kata."""

 return len(words)
```

```
def top_words(words, n=5):

 """Mengambil n kata yang paling sering muncul."""

 counter = Counter(words)

 return counter.most_common(n)
```

```
def count_vowels_consonants(text):

 """Menghitung jumlah huruf vokal dan konsonan."""

 vowels = "aeiouAEIOU"

 vowel_count = 0

 consonant_count = 0

 for char in text:

 if char.isalpha():

 if char in vowels:
```

```

 vowel_count += 1

 else:

 consonant_count += 1

 return vowel_count, consonant_count

def generate_bar_chart(top_words_list):

 """Membuat grafik ASCII."""

 chart = ""

 for word, count in top_words_list:

 chart += f"{word:<15} {'#' * count}\n"

 return chart

analyzer.py:

from utils import (

 read_file,

 clean_words,

 count_lines,

 count_words,

 top_words,

 count_vowels_consonants,

 generate_bar_chart

)

def analyze_text(input_file="input.txt", output_file="report.txt"):

 # Baca file

 text = read_file(input_file)

```

```

Proses kata

words = clean_words(text)

Statistik

total_lines = count_lines(text)

total_words = count_words(words)

top5 = top_words(words, 5)

vowels, consonants = count_vowels_consonants(text)

Grafik

chart = generate_bar_chart(top5)

Susun laporan

report = f"""

===== LAPORAN ANALISIS TEKS =====

Jumlah Baris : {total_lines}

Jumlah Kata : {total_words}

Jumlah Vokal : {vowels}

Jumlah Konsonan : {consonants}

5 Kata Terbanyak:

"""

for word, count in top5:

 report += f"- {word}: {count} kali\n"

report += "\nGrafik Frekuensi Kata:\n"

report += chart

```

```
Simpan ke file

with open(output_file, "w", encoding="utf-8") as file:

 file.write(report)

Tampilkan ke layar

print(report)

print(f"\nLaporan berhasil disimpan ke '{output_file}')
```

main.py:

```
from analyzer import analyze_text
```

```
def main():

 try:

 analyze_text("input.txt", "report.txt")

 except FileNotFoundError:

 print("File input.txt tidak ditemukan.")

 except Exception as e:

 print("Terjadi error:", e)
```

```
if __name__ == "__main__":

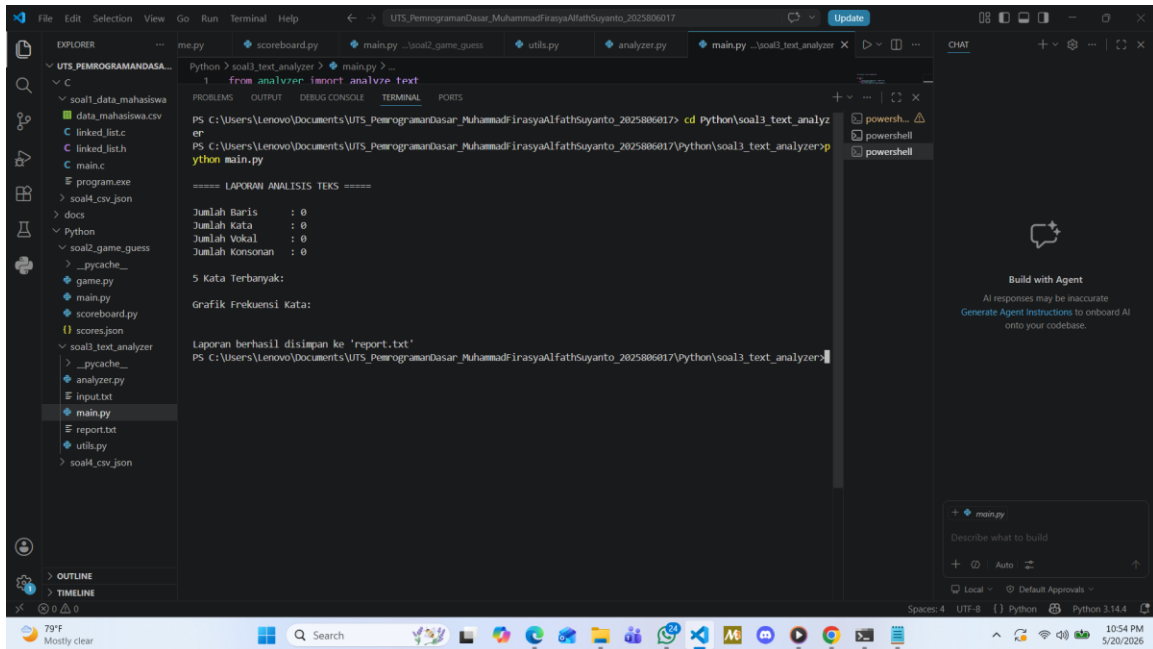
 main()
```

Instruksi run:

Cd Python\soal3\_text\_analyzer

Python main.py

Hasil output:



## 7. Soal 4 – Konversi CSV ke JSON (Python)

File: Python/csv\_to\_json.py

Convert.py:

```
import csv
```

```
import json
```

```
def csv_to_json():
```

```
 data = []
```

```
 try:
```

```
 # Membaca file CSV
```

```
 with open("data_mahasiswa.csv", "r", encoding="utf-8") as csv_file:
```

```
 reader = csv.DictReader(csv_file)
```

```
 for row in reader:
```

```
 mahasiswa = {
```

```

 "nama": row["Nama"],

 "nim": row["NIM"],

 "nilai_akhir": float(row["NilaiAkhir"]),

 "mutu": row["Mutu"]

 }

 data.append(mahasiswa)

Hitung rata-rata nilai akhir

if data:

 rata_rata = sum(m["nilai_akhir"] for m in data) / len(data)

 print(f"Rata-rata nilai akhir: {rata_rata:.2f}")

else:

 print("Data kosong.")

Tampilkan data

print("\n=== DATA MAHASISWA ===")

for mhs in data:

 print(

 f"{mhs['nama']} | "

 f"{mhs['nim']} | "

 f"{mhs['nilai_akhir']:.2f} | "

 f"{mhs['mutu']}"

)

Simpan ke JSON

with open("data_mahasiswa.json", "w", encoding="utf-8") as json_file:

 json.dump(data, json_file, indent=4)

print("\nData berhasil dikonversi ke data_mahasiswa.json")

```

```
except FileNotFoundError:

 print("File data_mahasiswa.csv tidak ditemukan.")

except Exception as e:

 print("Terjadi kesalahan:", e)
```

```
if __name__ == "__main__":

 csv_to_json()
```

```
data_mahasiswa.json:
```

```
[

 {

 "nama": "Rina",

 "nim": "2310001",

 "nilai_akhir": 85.5,

 "mutu": "A"

 },

 {

 "nama": "Doni",

 "nim": "2310002",

 "nilai_akhir": 61.5,

 "mutu": "C"

 },

 {

 "nama": "Siti",

 "nim": "2310003",

 "nilai_akhir": 91.4,

 "mutu": "A"

 },

 {
```



```

 "nama": "Budi",

 "nim": "2310004",

 "nilai_akhir": 75.5,

 "mutu": "B"

 }

]

```

Instruksi run:

Cd Python\soal4\_csv\_json

Python convert.py

Hasil output:

```

def csv_to_json():
 print("\n=== DATA MAHASISWA ===")
 for mhs in data:
 print(
 f"[{mhs['nama']} | "
 f"{mhs['nim']} | "
 f"{mhs['nilai_akhir']:.2f} | "
 f"{mhs['mutu']}]"
)

 # Simpan ke JSON
 with open("data_mahasiswa.json", "w", encoding="utf-8") as json_file:
 json.dump(data, json_file, indent=4)

 print("\nData berhasil dikonversi ke data_mahasiswa.json")

except FileNotFoundError:
 print("File data_mahasiswa.csv tidak ditemukan.")
except Exception as e:
 print("Terjadi kesalahan:", e)

if __name__ == "__main__":
 csv_to_json()

```

```

PS C:\Users\Lenovo\Documents\UTS_PemrogramanDasar_MuhammadFasyaAlfathSuyanto_2025806017> cd Python\soal4_csv_json
PS C:\Users\Lenovo\Documents\UTS_PemrogramanDasar_MuhammadFasyaAlfathSuyanto_2025806017\Python\soal4_csv_json> python convert.py
Rata-rata nilai akhir: 85.50

=== DATA MAHASISWA ===
Andi | 2024806033 | 85.50 | A
Data berhasil dikonversi ke data_mahasiswa.json

```

## 8. Soal 4 – Konversi CSV ke JSON (Python)

File: Python/csv\_to\_json.py

Data\_mahasiswa.json:

import csv

```
import json
```

```
def convert_csv_to_json():
```

```
 data = []
```

```
 try:
```

```
 # Membaca file CSV hasil dari program C (Soal 1)
```

```
 with open("data_mahasiswa.csv", "r", encoding="utf-8") as csv_file:
```

```
 reader = csv.DictReader(csv_file)
```

```
 for row in reader:
```

```
 mahasiswa = {
```

```
 "nama": row["Nama"],
```

```
 "nim": row["NIM"],
```

```
 "tugas": float(row["Tugas"]),
```

```
 "uts": float(row["UTS"]),
```

```
 "uas": float(row["UAS"]),
```

```
 "nilai_akhir": float(row["NilaiAkhir"]),
```

```
 "mutu": row["Mutu"]
```

```
 }
```

```
 data.append(mahasiswa)
```

```
 # Menghitung rata-rata nilai akhir
```

```
 if data:
```

```
 rata_rata = sum(m["nilai_akhir"] for m in data) / len(data)
```

```

 print(f"Rata-rata nilai akhir: {rata_rata:.2f}")
 else:
 print("Data kosong.")

Menampilkan data
print("\n=== DATA MAHASISWA ===")

for m in data:
 print(
 f"{m['nama']} | "
 f"{m['nim']} | "
 f"{m['nilai_akhir']:.2f} | "
 f"{m['mutu']}"
)

Menyimpan ke file JSON
with open("data_mahasiswa.json", "w", encoding="utf-8") as json_file:
 json.dump(data, json_file, indent=4)

print("\nData berhasil dikonversi ke data_mahasiswa.json")

except FileNotFoundError:
 print("File data_mahasiswa.csv tidak ditemukan.")
 print("Jalankan Soal 1 terlebih dahulu dan pilih menu 'Simpan ke CSV'.")
except Exception as e:
 print("Terjadi kesalahan:", e)

```

```
if __name__ == "__main__":
```

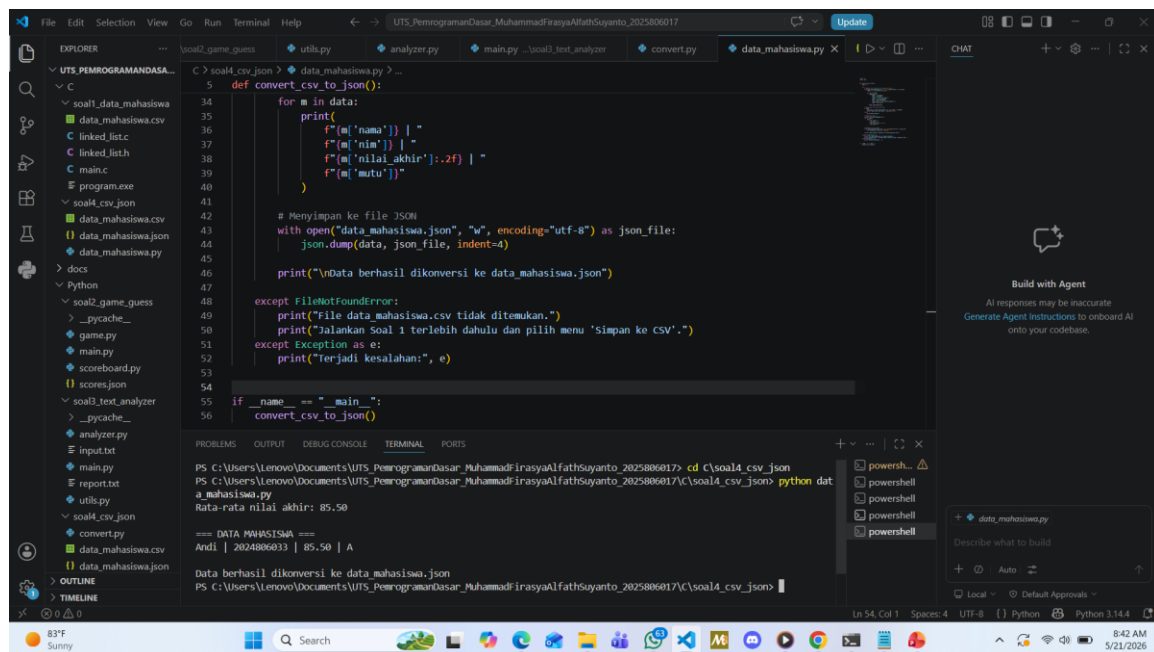
```
 convert_csv_to_json()
```

Instruksi run:

Cd C\soal4\_csv\_json

Python data\_mahasiswa.py

Hasil output:



The screenshot shows a Visual Studio Code editor with a Python file named `data_mahasiswa.py` open. The code defines a function `convert_csv_to_json()` that reads a CSV file and converts its contents to a JSON file. The script includes error handling for `FileNotFoundError` and `Exception`. The terminal output shows the command `cd C:\soal4_csv_json` and `python data_mahasiswa.py` being executed. The output displays the average final score: `Rata-rata nilai akhir: 85.50` and the converted JSON data: `== DATA MAHASISWA ==` and `Andi | 2024080933 | 85.50 | A`. The status bar at the bottom indicates the file is in UTF-8 encoding and is being edited with Python 3.14.4.

## 9. Bonus Opsional

Dokumen ini juga dapat dikembangkan dengan unit test (pytest), sorting data mahasiswa di C, dan penggunaan argparse pada Python.